

Brain Tumour Segmentation using U-Net and U-Net++ on the LGG MRI Dataset

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Abstract

In this study we compare binary tumor segmentation of the FLAIR modality (lower grade glioma) in brain MRI employing two from-scratch architectures based on encoders-decoders U-Net and U-Net++. On the LGG dataset we compare the two architectures under identical conditions by means of identical training pipeline, conduct ablations regarding the loss function, data augmentation and tumour slices sampling, and measure Dice, IoU and per-pixel precision-recall-F1, in addition to post-processing and visualization (Seg-Grad-CAM). We find that our most significant finding is that the choice of loss function was more crucial than the choice of architecture: binary cross-entropy reached a tumour-slice Dice of 0.655 for U-Net and 0.625 for U-Net++, far above the Dice+BCE combination (0.404 in U-Net and 0.482 in U-Net++); at their best configurations the two networks were very close, within run-to-run variation. The weaker Dice+BCE was boosted to 0.589 by tumour slices oversampling, while no noticeable effect was observed by data augmentation in any single run.

1 Motivation

Low-grade gliomas are among the most common types of primary brain cancer, and tumour region segmentation on an MRI is essential for diagnostics, treatment planning, and assessing treatment efficacy. At present, tumour segmentation is done manually by specialists; it is time-consuming and error-prone due to possible inter-observer variability. Automatic segmentation is faster and more accurate, but it is complicated due to high tumour heterogeneity, blurry boundaries, and relatively small size of the abnormal region. It is a classical computer vision task and machine learning is quite applicable here because convolutional neural networks learn multi-level features, edges and textures in the lower levels, shapes and regions in the upper level which surpass hand-crafted features when applied to medical images. U-Net has been successfully applied to biomedical image segmentation, while U-Net++ has been proposed with nested and dense skip connections to solve the issue with boundary representation. This project aims to validate the aforementioned hypothesis experimentally rather than trust it blindfolded: does the use of nested skip connections in U-Net++ provide higher segmentation accuracy in comparison with a plain U-Net model? Does the choice of loss function, data augmentation, and class imbalance handling have an impact on performance?

2 Related Work

The U-Net of Ronneberger et al. [1] introduced a symmetric encoder-decoder with skip connections that copy encoder features directly to the decoder, enabling precise localisation from few training images; it remains the reference architecture for biomedical segmentation. U-Net++ [2] re-designed these skips as nested, densely connected pathways, with intermediate convolutional

nodes that progressively refine encoder features before they are fused in the decoder, with the stated goal of reducing the semantic gap between encoder and decoder representations. For brain tumour data, the LGG MRI collection of Buda et al. [3] provides FLAIR-abnormality masks for lower-grade glioma patients and is a widely used benchmark for binary tumour segmentation, with reported U-Net Dice scores typically in the 0.8–0.9 range on all slices. For interpretability, Seg-Grad-CAM [4] extends gradient-weighted class activation mapping to dense prediction, highlighting the image regions that most influence a segmentation decision.

3 Objectives

- Implement U-Net and U-Net++ from scratch in PyTorch.
- Train both architectures with an identical pipeline for a fair comparison.
- Evaluate with Dice, IoU and pixel precision/recall/F1.
- Ablate the loss function (Dice+BCE vs. BCE), data augmentation (on/off) and tumour-slice oversampling (on/off).
- Provide qualitative analysis via prediction overlays and Seg-Grad-CAM.
- Apply morphological post-processing and quantify its effect.

4 Dataset

We use the LGG MRI Segmentation dataset [3], comprising pre-operative brain MRI from 110 lower-grade glioma patients, distributed as 3,929 axial slices in three-channel `.tif` format, each with a binary mask of the FLAIR abnormality (tumour vs. background). During loading we detected and removed a duplicated copy of the data (a recursive search initially returned twice the expected number of pairs); de-duplicating by filename restored the correct 3,929 slices.

To prevent data leakage we split at the patient level into 76/17/17 patients for training/validation/test, corresponding to 2,839/557/533 slices. Because multiple adjacent slices of the same patient are highly correlated, a naive slice-level split would place near-identical images in both training and test sets and inflate the scores; patient-level splitting avoids this. A large fraction of slices contain no tumour at all, which has two consequences we address below: it motivates a tumour-slice oversampling experiment, and it requires reporting Dice on tumour-containing slices separately, since a correct empty prediction scores a perfect Dice of 1 and would otherwise inflate the all-slice average.

5 Dataset Exploration

Before modelling, we examined the dataset to understand its difficulty. Figure 1 shows representative slices with the tumour mask overlaid: the FLAIR abnormality appears as a comparatively bright, irregular region, sometimes compact and well-defined and sometimes diffuse with fuzzy borders, and it varies considerably in size and location across patients. Figure 2 summarises three properties that shaped our design choices. First, the majority of slices contain no tumour at all—only 1,373 of the 3,929 slices (34.9%) are tumour-positive, while 65.1% are empty. So training naïvely on every slice biases the model toward predicting empty masks; this motivates the tumour-slice oversampling experiment. Second, even on the slices that do contain tumour, the lesion occupies on average only about 3% of the pixels, which is why we report Dice on tumour-containing slices separately rather than letting easy empty slices inflate the average. Third, the number of tumour-positive slices varies widely across the 110 patients, which is why

the train/validation/test split is made at the patient level: splitting by slice would scatter correlated neighbouring slices of the same patient across the sets and inflate the apparent scores.

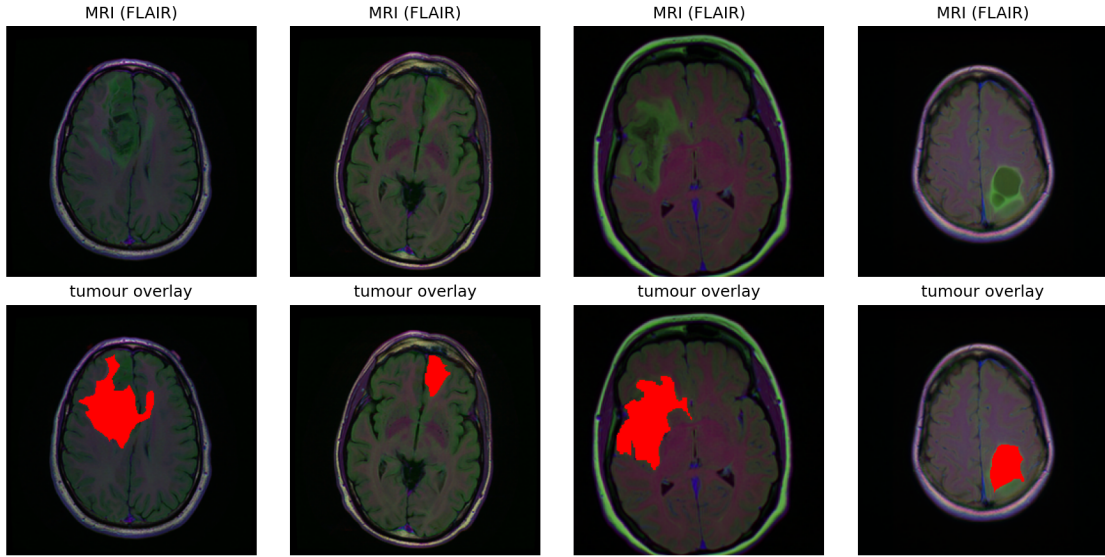


Figure 1: Example MRI slices (top) and the same slices with the ground-truth tumour mask overlaid in red (bottom).

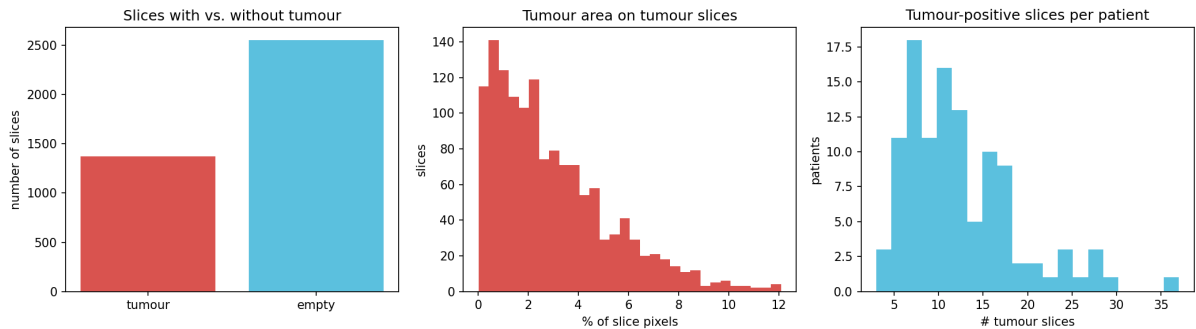


Figure 2: Dataset statistics. Left: number of slices with vs. without tumour. Centre: distribution of tumour area (percentage of slice pixels) over tumour-containing slices. Right: number of tumour-positive slices per patient.

6 Methodology

6.1 Pre-processing

Each slice and its mask were resized to 128×128 pixels and the image was scaled to $[0, 1]$; the mask was binarised at 0.5. We chose 128×128 rather than the native 256×256 because it reduces the computational cost roughly fourfold with negligible loss in Dice on this dataset, where the abnormal region remains clearly visible at the lower resolution. The patient-level split described above was applied before constructing the data loaders.

6.2 Architectures

Both architectures have been created from scratch and employ the same double-convolution block (two 3×3 convolutions with batch normalisation and ReLU after each). The U-Net

consists of four levels of encoders and decoders with max-pooling for downsampling, transposed convolutions for upsampling, and a single skip connection at each level with concatenation of the corresponding encoder feature map into the decoder. It has about 7.8M of parameters. The U-Net++ replaces each skip with a hierarchical set of nodes $x^{i,j}$: instead of passing the encoder features to the decoder as they are, it passes them through intermediate convolutional blocks which are densely connected and progressively upsampled, so that decoder features are merged with the repeatedly transformed encoder features. The nesting is done to fill the semantic gap between the low-level features from the encoder and high-level features from the decoder; this increases the number of parameters to approximately 9.2M.

6.3 Loss functions

We compare two losses. Binary cross-entropy (BCE) evaluates every pixel independently:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (1)$$

The soft Dice loss instead measures regional overlap and is, in principle, less sensitive to class imbalance because it is a ratio rather than a per-pixel sum:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i \hat{y}_i y_i + \varepsilon}{\sum_i \hat{y}_i + \sum_i y_i + \varepsilon}. \quad (2)$$

Our hybrid loss is $\mathcal{L} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 \mathcal{L}_{\text{Dice}}$. We treat the choice between the hybrid and BCE alone as an ablation rather than assuming the hybrid is superior, precisely because the relative benefit of a Dice term depends on how severe the class imbalance is.

6.4 Training

Both models were trained for 40 epochs with the Adam optimiser (learning rate 10^{-3}), a cosine-annealing schedule and mixed-precision arithmetic. We retained the checkpoint with the best all-slice validation Dice, which is a smooth and stable selection criterion; the tumour-slice validation Dice is noisier from epoch to epoch (the validation set contains only 17 patients) and selecting on it directly risked locking onto an early, unrepresentative spike. The tumour-slice Dice is nonetheless reported as the primary test metric.

6.5 Evaluation metrics

The Dice coefficient and Intersection-over-Union are defined as

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}, \quad \text{IoU} = \frac{|P \cap G|}{|P \cup G|}, \quad (3)$$

for predicted mask P and ground-truth mask G . We report Dice over tumour-containing slices as the headline metric, all-slice Dice and IoU, and pixel-level precision, recall and F1 for the tumour class.

6.6 Post-processing

At inference, a light morphological clean-up fills small holes and removes connected components below a size threshold, suppressing isolated false-positive specks while leaving the main predicted region intact.

7 Results

Table 1 reports all seven configurations on the held-out test set, sorted by tumour-slice Dice. Three patterns stand out: the loss function is the dominant factor; the two architectures are very close at their best setting; and tumour-slice oversampling clearly helps the weak Dice+BCE baseline, while augmentation does not.

Table 1: Test-set results for all configurations, sorted by tumour-slice Dice. Aug = augmentation; OS = tumour-slice oversampling.

Model	Loss	Aug	Dice (tumour)	Dice (all)	IoU	F1
U-Net	BCE	on	0.655	0.832	0.800	0.791
U-Net++	BCE	on	0.625	0.825	0.797	0.792
U-Net	Dice+BCE	on (OS)	0.589	0.794	0.763	0.725
U-Net++	Dice+BCE	off	0.508	0.799	0.772	0.705
U-Net++	Dice+BCE	on	0.482	0.776	0.750	0.687
U-Net	Dice+BCE	off	0.439	0.767	0.743	0.643
U-Net	Dice+BCE	on	0.404	0.764	0.738	0.629

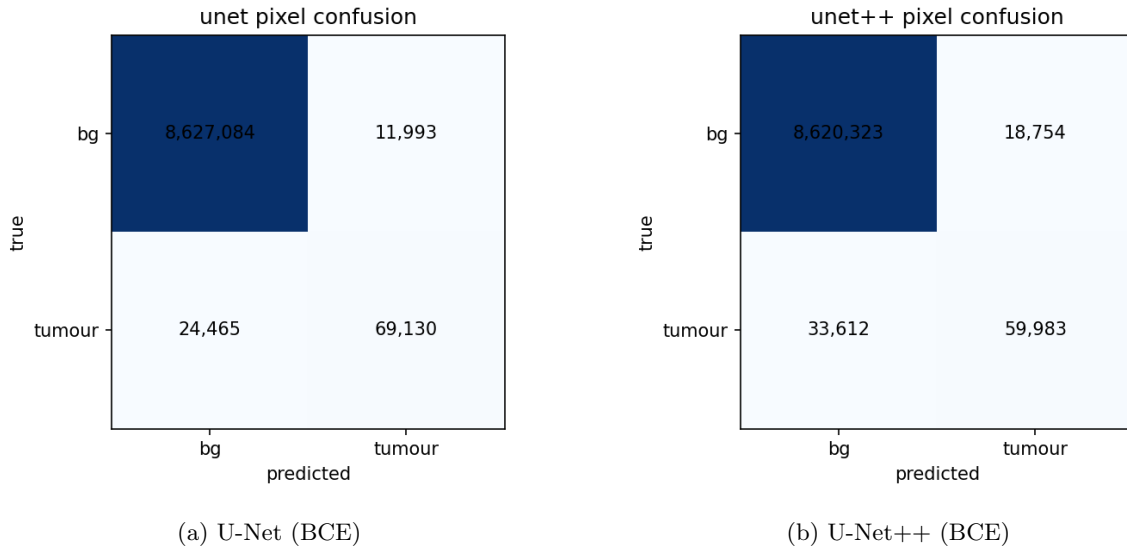


Figure 3: Pixel-level confusion matrices for the two best models. Most error is background/tumour confusion at the lesion boundary; both models recover the majority of tumour pixels.

7.1 Loss function

The loss function produced the largest effect in the entire study, and the effect was consistent across both architectures. Replacing the Dice+BCE hybrid with plain BCE raised the tumour-slice Dice from 0.404 to 0.655 for U-Net and from 0.482 to 0.625 for U-Net++ an improvement of roughly 0.14–0.25 in each case. Because the same effect appears on two different architectures, it is unlikely to be a chance artefact of a single run.

At first glance, this seems to be counter-intuitive: Dice loss is traditionally used for imbalance. The reason behind this choice is *the level of imbalance*. In this particular dataset, the area where the lesion occurs is relatively big and high contrast on FLAIR scan, but the background is not too much of an area in comparison with tumor-containing slices. In such a situation, BCE receives a solid and conditioned gradient from the lesion part and trains steadily, while the soft-Dice is just a ratio with noised gradients which prevent convergence when added equally.

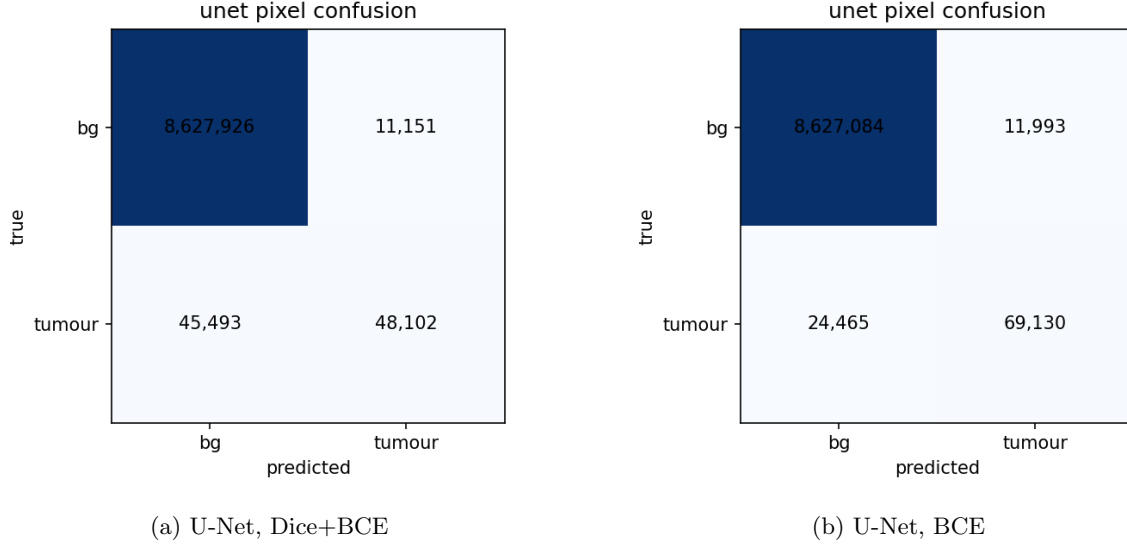


Figure 4: Effect of the loss function on U-Net. Switching from the Dice+BCE hybrid to plain BCE markedly increases the number of correctly recovered tumour pixels.

7.2 Architecture comparison

At their best operating point the two architectures are very close: U-Net with BCE reaches a tumour-slice Dice of 0.655 and U-Net++ reaches 0.625. U-Net is in fact marginally ahead on tumour-slice Dice, all-slice Dice (0.832 vs. 0.825) and IoU (0.800 vs. 0.797), while U-Net++ is marginally ahead on F1 (0.792 vs. 0.791); these differences are well within run-to-run variation. Under the relatively weaker Dice+BCE loss function, U-Net++ performs better than U-Net (0.482 vs. 0.404); nevertheless, they perform much worse than the versions of U-Net++ and U-Net that utilize BCE loss function; the performance difference decreases under a more efficient loss function.

In reality, the densely connected skip connections feature in U-Net++ does not offer an observable gain at the optimal operating point in terms of accuracy, with increased parameters from 7.8M to 9.2M and training time almost tripled per epoch due to additional architectural complexity. Therefore, it seems unnecessary to incorporate this extra complexity into this architecture for this particular dataset, as the accuracy gain is less than the accuracy variation between experiments, while the computation cost has significantly increased.

7.3 Class-imbalance handling: oversampling

Tumour-slice oversampling produced a clear, positive effect. Starting from the weak Dice+BCE baseline (0.404), drawing tumour-containing slices more frequently during training raised the tumour-slice Dice to 0.589. Without oversampling, a large share of each training batch consists of tumour-free slices, biasing the network towards predicting empty masks; oversampling exposes the model to active lesions more often, so it learns the visual signature of abnormal FLAIR tissue more effectively. Together with the loss-function result, this gives two complementary levers for handling imbalance, changing what the loss optimises, and changing what the model sees, both of which improve tumour recovery.

7.4 Augmentation

Simple flip-and-rotation augmentation did not improve performance on this dataset; if anything it slightly reduced it. With augmentation turned off the tumour-slice Dice was higher for both models (0.439 vs. 0.404 for U-Net; 0.508 vs. 0.482 for U-Net++), and the three-seed repetition

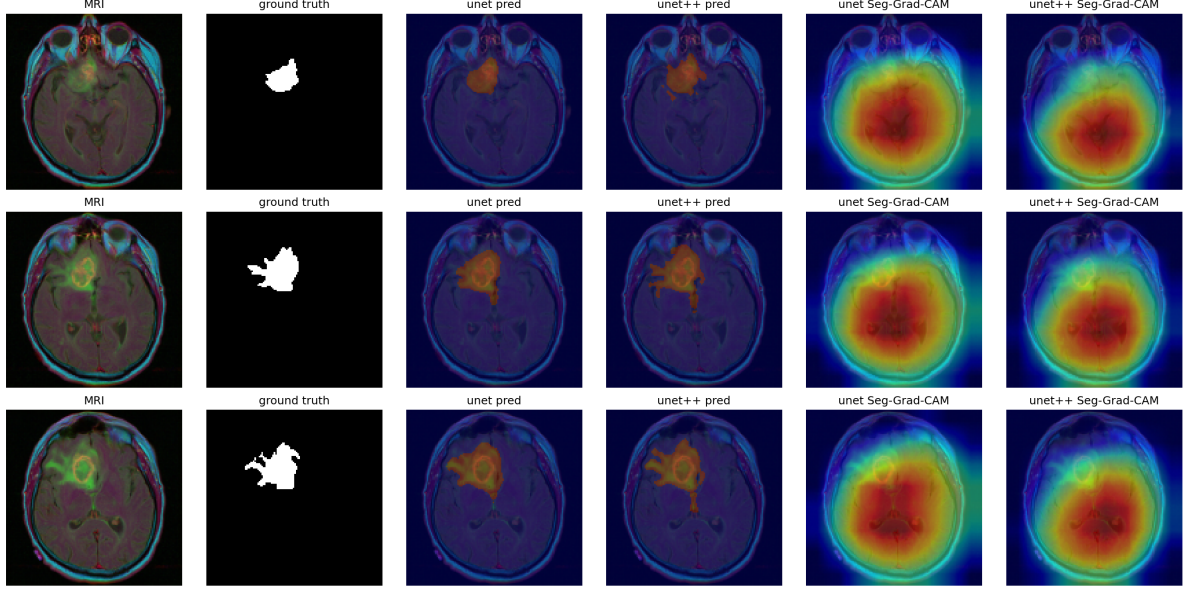


Figure 5: U-Net vs. U-Net++ predictions and Seg-Grad-CAM saliency maps on tumour-containing slices. The saliency concentrates on the abnormal FLAIR region for both models, with no qualitative advantage for the nested architecture.

in the Limitations section shows the same direction with non-overlapping ranges (0.522 ± 0.022 off vs. 0.430 ± 0.030 on for U-Net Dice+BCE). We attribute this to the dataset already being large and varied enough (2,839 training slices) that simple spatial augmentation mainly added noise rather than useful variation; the effect is small and specific to this dataset and these augmentations.

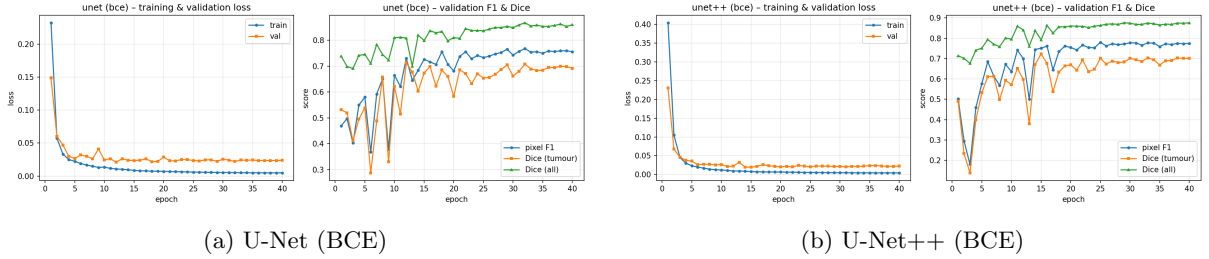


Figure 6: Training/validation loss (left of each pair) and validation F1/Dice (right of each pair) over epochs. Loss decreases steadily while F1 and Dice rise and plateau; the all-slice Dice is smooth, whereas the tumour-slice curve is noisier, which is why checkpoints were selected on the former.

7.5 Post-processing

Morphological post-processing gave a small, consistent improvement and never degraded performance. For the best U-Net the all-slice Dice rose from 0.832 to 0.850, and similar small gains appear across configurations. The effect is limited because the trained models already produce fairly clean masks; the clean-up mainly removes isolated false-positive specks rather than recovering missed tumour, which is the expected behaviour of a size-and-hole filter.

8 Discussion

Taken together, the experiments support a single clear conclusion: on this task the choice of loss function influenced tumour segmentation far more than the choice of architecture. Plain BCE

substantially outperformed the Dice+BCE hybrid on both networks, while U-Net and U-Net++ were very close at their best setting (0.655 vs. 0.625). This changes the traditional perspective on U-Net++: there is no guarantee for the benefits of its nested skip connections, and in a mildly unbalanced and highly contrasted modality like FLAIR, it delivered none while costing three times more to train. A three-seed repetition of this comparison (Table 2) confirms the point: the U-Net and U-Net++ tumour-Dice ranges overlap, so the two architectures are statistically indistinguishable on this task.

The imbalance findings are no less revealing either. Both approaches which directly address imbalance over-representing tumour slices and, by inference, the easier optimization of BCE enhances tumour recovery, but the architecture change does not. The Seg-Grad-CAM maps in Figure 5 show that both models focus their attention on the abnormal FLAIR region, and the qualitative overlays in Figure 7 indicate that the remaining errors are concentrated on small, low-contrast lesions and on diffuse boundaries, exactly the cases where Dice scores of around 0.65 rather than 0.9 are to be expected.

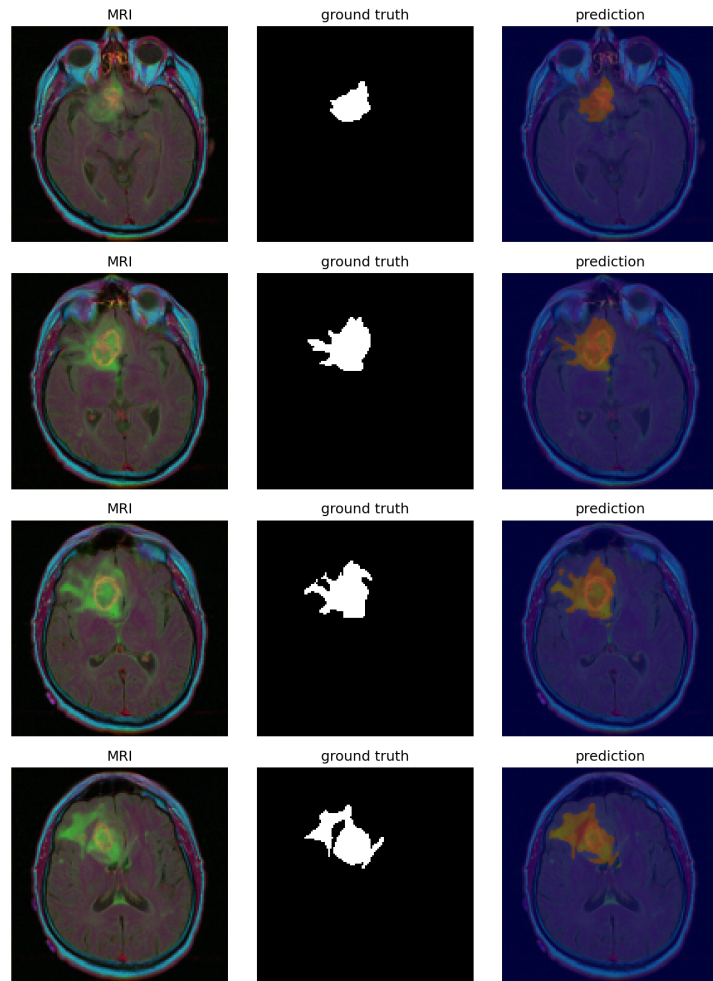


Figure 7: Qualitative results for U-Net++ (BCE): MRI input, ground-truth mask and predicted overlay. Errors concentrate on small or low-contrast lesions and on fuzzy boundaries.

9 Limitations and Future Work

To check whether the small gaps are real, we re-ran the two close comparisons over three random seeds (42, 123, 999) and report mean $\pm$ standard deviation of the tumour-slice Dice. For the architecture comparison, U-Net scored 0.629 ± 0.044 and U-Net++ 0.656 ± 0.012 : the two ranges

overlap, confirming that the difference between the architectures is within run-to-run variation rather than a genuine advantage of either. For augmentation, U-Net with Dice+BCE scored 0.430 ± 0.030 with augmentation on and 0.522 ± 0.022 with it off, so the augmentation effect is likewise small and noisy.

Table 2: Multi-seed robustness on the two close comparisons: tumour-slice Dice as mean $\pm$ standard deviation over three seeds (42, 123, 999).

Model	Loss	Aug	Dice (tumour)
U-Net	BCE	on	0.629 ± 0.044
U-Net++	BCE	on	0.656 ± 0.012
U-Net	Dice+BCE	on	0.430 ± 0.030
U-Net	Dice+BCE	off	0.522 ± 0.022

Two limitations remain. The tumour-slice validation metric was noisy, which is why checkpoints were selected on the smoother all-slice Dice. The pipeline also operates on 2-D slices and therefore ignores 3-D context that a volumetric model could exploit. Future work could evaluate a focal-Tversky loss, a pretrained encoder, test-time augmentation, and a 3-D U-Net variant.

10 Conclusion

We implemented U-Net and U-Net++ from scratch and compared them for FLAIR abnormality segmentation on the LGG dataset under a controlled set of ablations. Our central result is that the loss function dominated: BCE reached a tumour-slice Dice of 0.655 for U-Net and 0.625 for U-Net++, well above the Dice+BCE hybrid, while the nested U-Net++ offered no measurable accuracy advantage over U-Net at three times the training cost. Tumour-slice oversampling further improved the weak baseline, augmentation had no clear effect, and post-processing gave a small consistent gain. The study illustrates that, for a moderately imbalanced and high-contrast modality, a careful choice of loss and sampling is more decisive than architectural complexity.

References

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